Terminal

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lelouch@fedora:~$ node /home/lelouch/Documentos/secante.js

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Ejercicio 1: ln(x) - 2x + 3 = 0
x0 = 1, x1 = 2
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(index)	Iteracion	x aproximado	f(x)	Error (%)
0	0	'1.000000000'	'1.000000e+0'	'---'
1	1	'2.000000000'	'-3.068528e-1'	'---'
2	2	'1.765197110'	'3.786814e-2'	'13.301794'
3	3	'1.790990578'	'7.877066e-4'	'1.440179'
4	4	'1.791538513'	'-2.271214e-6'	'0.030585'

```
Estado: CONVERGENCIA ALCANZADA
Raíz aproximada: 1.7915385132590194
Error final: 0.030585 %

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Ejercicio 2: x^3 - 3x - 4 = 0
x0 = 5, x1 = 7
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(index)	Iteracion	x aproximado	f(x)	Error (%)
0	0	'5.000000000'	'1.060000e+2'	'---'
1	1	'7.000000000'	'3.180000e+2'	'---'
2	2	'4.000000000'	'4.800000e+1'	'75.000000'
3	3	'3.466666667'	'2.726163e+1'	'15.384615'
4	4	'2.765573208'	'8.855477e+0'	'25.350747'
5	5	'2.428266587'	'3.033422e+0'	'13.890840'
6	6	'2.252522210'	'6.714074e-1'	'7.802115'
7	7	'2.202566524'	'7.760984e-2'	'2.268067'
8	8	'2.196037273'	'2.452964e-3'	'0.297320'
9	9	'2.195824172'	'9.474893e-6'	'0.009705'

```
Estado: CONVERGENCIA ALCANZADA
Raíz aproximada: 2.1958241718698757
Error final: 0.009705 %
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Ejercicio 3: x^2 - x + 6 = 0
Discriminante = -23
La ecuación no tiene raíces reales.
Se ejecutará la secante para observar su comportamiento.

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Ejercicio 3: x^2 - x + 6 = 0
x0 = 1, x1 = 2
====================================
```

(index)	Iteracion	x aproximado	f(x)	Error (%)
0	0	'1.000000000'	'6.000000e+0'	'---'
1	1	'2.000000000'	'8.000000e+0'	'---'
2	2	'-2.000000000'	'1.200000e+1'	'200.000000'
3	3	'10.000000000'	'9.600000e+1'	'120.000000'
4	4	'-3.714285714'	'2.351020e+1'	'369.230769'
5	5	'-8.162162162'	'8.078305e+1'	'54.493851'
6	6	'-1.888455772'	'1.145472e+1'	'332.213573'
7	7	'-0.851887406'	'7.577600e+0'	'121.679034'
8	8	'1.174022837'	'6.204307e+0'	'172.561400'
9	9	'10.326746067'	'1.023149e+2'	'88.631241'
10	10	'0.583179744'	'5.756919e+0'	'1670.765562'

101	101	-0.095708927	6.104869e+0	1735.022562

```
Estado: NO SE ALCANZÓ LA TOLERANCIA
Última aproximación: -0.09570892651479657
Revise la existencia de raíces y los puntos iniciales.

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Ejercicio 4: 2 sin(x) - x = 0
x0 = 1, x1 = 2
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(index)	Iteracion	x aproximado	f(x)	Error (%)
0	0	'1.000000000'	'6.829420e-1'	'---'
1	1	'2.000000000'	'-1.814051e-1'	'---'
2	2	'1.790124658'	'1.619630e-1'	'11.724063'
3	3	'1.889120548'	'1.040191e-2'	'5.240316'
4	4	'1.895914816'	'-6.890454e-4'	'0.358364'
5	5	'1.895492710'	'2.550877e-6'	'0.022269'

```
Estado: CONVERGENCIA ALCANZADA
Raíz aproximada: 1.8954927097634646
Error final: 0.022269 %
```